Welcome to Chemistry 370

This is a one-semester physical chemistry course intended for chemical engineering majors (or others with a prior knowledge of chemical thermodynamics). The subject matter of this course is primarily quantum theory and its applications to atomic and molecular spectroscopy and structure.

Learning Outcomes and Objectives

1. Give examples that demonstrate the fundamental principles in quantum mechanics.
    - 1.1. Energy quantization, particle wave duality
    - 1.2. Operators, eigenfunctions, eigenvalues
    - 1.3. Schrödinger equation, wavefunctions
    - 1.4. Hamiltonian and energy

2. Apply simple quantum mechanical models such as particle-in-a-box to atomic and molecular structures to predict wave functions and calculate energy levels.
    - 2.1. Particle in a box
    - 2.2. Particle on a ring
    - 2.3. Harmonic oscillator
    - 2.4. Tunneling through barriers
    - 2.5. Particle on a sphere, 3D particle in a box

3. Interpret simple optical spectroscopy of atoms and molecules based on quantum mechanical principles.
    - 3.1. Atomic and molecular orbitals
    - 3.2. Molecular orbitals and chemical bonding
    - 3.3. Absorption, emission, vibrational and rotational spectroscopy; selection rules

4. Other topics
    - 4.1. Spin
    - 4.2. Permutation symmetry/ Quantum statistics
    - 4.3. Entanglement, Bell tests

Lectures MWF 11:30 PM -12:20 PM

In-class lectures in Wetherill Lab of Chemistry 172

Web Sites Bright Space
Course Page: https://purdue.brightspace.com/d2l/home/1358356

Bright Space will be used to distribute course materials, including homework assignments, any slides, and exam related information. All homework will be submitted on Bright Space. Most importantly, you can use Bright Space to post questions, answers, and notes (either anonymously or not), to help make this class work better for you.

Homework	Posted on Bright Space.
Midterm exam	Oct 8, 2025 (tentative)
Final exam	TBD
Professor	Lee R. Liu, leeliu@purdue.edu
Teaching assistants	Trevor Michael Du, du349@purdue.edu Shahed Haghiri, shaghiri@purdue.edu
Textbook	Introduction to Quantum Mechanics by David J. Griffiths (Cambridge University Press, 2018) ISBN 978-1107189638

Office hours

	Trevor	Shahed	Lee
Date	Mondays	Tuesdays	Every other Friday starting Sept 12: **September:** 12, 26 **October:** 10, 24 **November:** 7, 21 **December:** 5
Time	3-4pm	3-4pm	3:30 -4:30 pm
Location	BRWN 1152, except for 9/8 which is booked for BRWN B106.	BRWN 1152, except for 9/9 and 9/30 which will be in WTHR 277.	BRWN 4171C

Homework
There is no better way to learn the material in this course than by doing homework problems. Learning in whatever way that suits you best is encouraged, whether that is through self-study, study groups, office hours, and/or consulting AI tools. However, the homework you hand in should be yours alone. To reinforce this, and to disincentivize handing in homework without understanding it, **full marks are given for each homework question that you make a good faith attempt at completing, regardless of whether you arrive at the correct answer (this will NOT be the case on midterms and exams; see below).** Answer keys will be released after homework due dates so you can check your work and prepare for midterms and exams. Selected

questions may be reviewed in class. You are permitted to use symbolic math software like Wolfram Alpha to aid in carrying out matrix multiplications, integrals, and derivatives. Cheating on any question will result in a grade of zero for the whole assignment. Each homework will be distributed every Wednesday after class and it will be due the following Wednesday at 11:59pm.

Please submit homework to Brightspace in **(i) pdf form**, **(ii) portrait orientation, (iii) right-side up**. Any other formats won't be graded and will receive zero. This helps your instructors streamline grading.

Late policy:
33% off if handed in Thursday
66% off if handed in Fri before 4pm
100% off if handed in Fri after 4pm (ie, you get zero)

This helps us grade everyone's homework early and give everyone feedback as quickly as possible.

Exams

Two in-class exams will be given during the semester. The midterm exam will include questions on any of the material covered in week 1-week 8 (approximately). The final exam will include questions on any of the material covered in week 9-week 16 (approximately). Exams are closed book and exam questions will follow a grading schemes similar to the homework answer keys.

Weighting of Assessments and Grading

In this class, grades reflect the sum of your achievement of learning outcomes throughout the session. At the end of the session, final grades will be calculated by weighing your percentages on all homework assignments and exams as follows:

Homework (throughout the session)	40%
Midterm exam	30%
Final exam	30%

Supplementary (not required) Textbooks on Reserve in the Chemistry Library

Title	Author(s)
Understanding Physical Chemistry	Ben-Amotz
Physical Chemistry	Moore
Physical Chemistry	McQuarrie & Simon
Physical Chemistry	Atkins & de Paula
Physical Chemistry	Castellan
Physical Chemistry	Engel & Reid
Molecular Theory of Gases and Liquids	Hirschfelder, Curtis & Bird

Academic Integrity:

Academic integrity is one of the highest values that Purdue University holds. Individuals are encouraged to alert university officials to potential breaches of this value by either emailing integrity@purdue.edu or by calling 765-494-8778. While information may be submitted anonymously, the more information is submitted the greater the opportunity for the university to investigate the concern.

Nondiscrimination Statement:

Purdue University is committed to maintaining a community that recognizes and values the inherent worth and dignity of every person; fosters tolerance, sensitivity, understanding, and mutual respect among its members; and encourages each individual to strive to reach his or her potential. In pursuit of its goal of academic excellence, the University seeks to develop and nurture diversity. The University believes that diversity among its many members strengthens the institution, stimulates creativity, promotes the exchange of ideas, and enriches campus life. See Purdue's full Nondiscrimination Policy Statement (https://www.purdue.edu/home/ea_eou_statement/)

Mental Health/Wellness Statement:

If you find yourself beginning to feel some stress, anxiety and/or feeling slightly overwhelmed, try Therapy Assistance Online (TAO), a new web and app-based mental health resource available courtesy of Purdue Counseling and Psychological Services (CAPS)**.** TAO is available to students, faculty, and staff at any time.

If you need support and information about options and resources, please contact or see the Office of the Dean of Students (ODOS). Call 765-494-1747.

If you find yourself struggling to find a healthy balance between academics, social life, stress, etc., sign up for free one-on-one virtual sessions with a Purdue Wellness Coach at RecWell. Student coaches can help you navigate through barriers and challenges toward your goals throughout the session. Sign up is free and can be done on BoilerConnect.

If you're struggling and need mental health services: Purdue University is committed to advancing the mental health and well-being of its students. If you or someone you know is feeling overwhelmed, depressed, and/or in need of mental health support, services are available. For help, such individuals should contact Counseling and Psychological Services (CAPS) at 765-494-6995 during and after hours, on weekends and holidays.

Basic Needs Security:

Any student who faces challenges securing their food or housing and believes this may affect their performance in the course is urged to contact the Dean of Students for support. There is no appointment needed, and Student Support Services is available to serve students 8 a.m.-5 p.m. Monday through Friday.